FM-1 · BORICUA RHYTHM · V1.0

COQUÍ



A Latin and Caribbean rhythm firmware for the M-VAVE FM-1: how it was built

A talk for the engineering team · November 2026

WHAT IT IS

A Latin-rhythm groovebox inside a pocket synth

An alternative firmware for the M-VAVE FM-1: it changes what the device does without changing the hardware. Installs over USB, plays out over USB, open source.

22	14	5	44	7

INSPIRATION

A night in Puerto Rico, in a 160 mm device

The coquí

Puerto Rico's symbol frog, which sings “co-quí” all night. It became the name, the mascot and the boot screen.

The music

Salsa, son, mambo, bomba, plena, merengue, bachata, cumbia, reggaeton and soca. The clave as the sense of time.

The reference

GR!MM, a community dark-ambient firmware: its structure, its tests and its hub were the model.

The FM-1 Firmware Hub lists more than 150 open firmwares for this device. The quality bar already existed; the

THE HARDWARE

The target: a chip with no FPU and 5.8 ms per audio block

Part	What it means
JieLi AC791N, 240 MHz	A pi32v2 core, with JieLi's own clang 4 compiler
No FPU	No floats: everything in fixed point (Q15), and no 64-bit division
96 KB of RAM	Plus a 336 KB pool for large buffers (reverb, delay)
1 MB of flash	~568 KB for the program; the rest for saves, samples and the updater
44.1 kHz, 256-frame blocks	Each audio block has 5.8 ms to be ready, or it crackles
240×240 screen, 27 keys	7 encoders, 14 buttons, LEDs, USB MIDI and audio, MIDI TRS

THE DEVICE

A pocket groovebox in aluminum

The M-VAVE FM-1: a 240×240 screen, 27 keys, 7 encoders and 4 knobs, USB and MIDI TRS. Cheap, closed from the factory — which is exactly what makes COQUÍ run on it.



FM-1_923 · COQUÍ

THE BASE

A fork of a lineage of open firmwares



Inherited: platform, sequencer, effects, saves, web editor and emulator — ~17k lines of already-tested C.

Changed: identity, saves, palette, screens, sounds, the VOICE source (now PERC) and GENERATE.

GPL-3.0: open source and credits kept; the upstream history became an initial commit.

ARCHITECTURE

It all happens in the audio interrupt, in three stages

1 · WHO PLAYS

The sequencer clock (96 ticks per quarter note) wakes, each tick:

- the track's steps
- GROOVE • groove.c
- the ARP lines • arp.c
- GENERATE • gen.c

2 · WHO SOUNDS

note_on() decides by the track's source:

- 6 shared voices: VA, FM, SAMPLE, WHEEL, PHYS, PHASE
- PERC: its own pool of 10 hits • perc.c

3 · THE OUTPUT

Per-track mixer and effects, tuned by AMBIENTE • amb.c:

- SPACE, DELAY, TAPE, GRAIN
- MASTER and DRIVE
- DAC and USB audio

22 synthesized drums, not a single sample

Each hit is a tone (sine, a second partial, a pitch drop) plus filtered noise (SVF), each decaying its own way.

Its own pool of 10 hits: the percussion never steals the 6 voices from the bass and the piano.

Choke groups: the open conga cuts the muted one, like a hand on the skin.

HUMAN: each hit a little late, a little softer and off-pitch, at random.



The SOUND screen of a PERC track

The rhythms are text: two bars per drum

```
CLAVE ....X...X.......X.....X.....X...

SLAP ....X..............X..........

CONGA ............x.x.............x.x.

BONGO x.o.x.o.x.o.x.o.x.o.x.o.x.o.x.o.
```



The same grid, on the FM-1 screen

X accent • **x** played • **o** soft • **g** ghost. 14 styles, with PARTS (split across tracks), FILL and FEEL (swing), played on the sequencer clock.

THE LINES · ARP.C

GR!MM's arpeggiator learned tumbao and montuno

```
TUMBAO ------5-----R---------5-----R--

MONTUNO ..5.3.5.R.5.3.5...5.3.5.O.5.3.O.

BACHATA R-----R-5-------R-----R-5-------
```

Degrees, not notes

One key is enough

The preset arms it

THE MOOD

Atmosphere at no cost: new rules over parts that already existed

GENERATE on the clave

Each sixteenth carries the weight of its place in a 2-3 clave. NOCHE, CALLE and FIESTA: 14, 51 and 112 notes in 32 s.

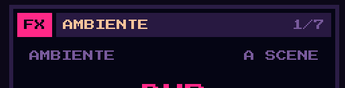
AMBIENTE

7 scenes (backyard, calle, dub, AM radio…) that tune the existing effects. Zero new DSP and zero new memory.

Night HOME

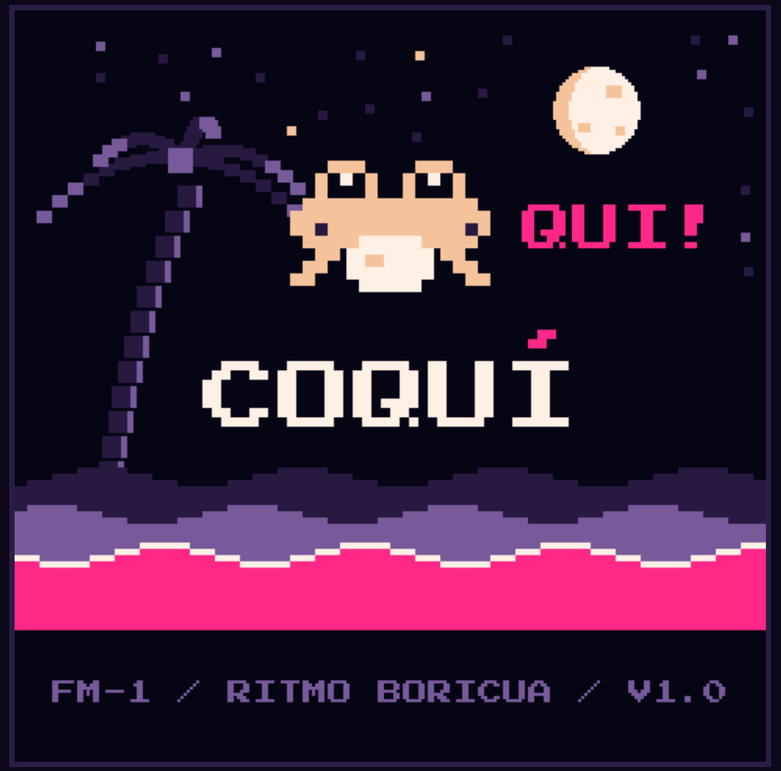
The coquí sings on the hits and sleeps in the silence; one conga with a neon flame per track.





Noche tropical: one palette, from the chip to the site

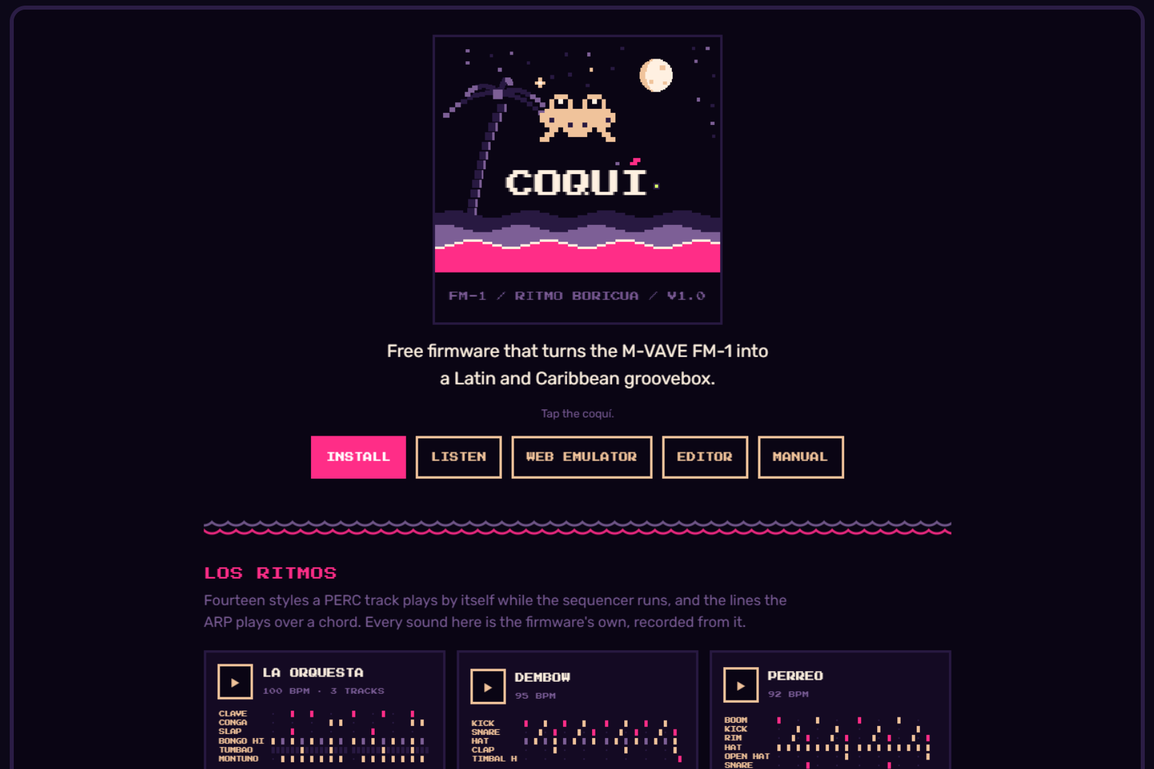
NIGHT #0C0818	DUSK #2B1D45	MAUVE #7D5F98
SAND #F2C49B	ROSA #FF2D8A	MANGO #FF9A1A
MAR #1CD8C4	LIMA #A6E82E	CREAM #FFF3E2



The pixel art is code: sprites as strings in the C, drawn on a 240×240 canvas. The same scene was ported to JavaScript on the hub, and this deck uses the same colors.

HUB · EMULATOR · EDITOR

The same C code runs on the FM-1, in the browser and in CI



Hub (GitHub Pages)

The audio is recorded from the firmware itself in host mode and turned into MP3 by a script.

Emulator (WebAssembly)

The firmware compiled with Emscripten; the test checks the sound and screens match the host, sample for sample.

CHALLENGES

What was hard, and how it was solved

Challenge	Solution
No FPU: the link failed on a 64-bit division	Fixed point everywhere; the decays redone with 32-bit divisions only
Buffer memory full (342 of 344 KB)	AMBIENTE as scenes over the existing effects; the choirs cut (–30 KB)
Only 6 voices for 4 tracks	Percussion with its own pool of 10 hits
The inherited save format with no room	Track from 52 to 56 bytes (version 9) and its own magics
New FM-1s ship with a different flash chip	Accept both JEDEC IDs in the app, the rescue and the loader
GR!MM's tests tied to its own sounds	Re-pointed to equivalents; references remade on purpose
Tools that only accepted GR!MM	Found by reading the code before the device: accept FM-1_923
WSL has no USB MIDI; rtmidi has no package on new Python	Install from Windows, with Python 3.12

QUALITY

Test off the device, so the device only confirms

18 host-mode suites: audio compared to references, screens, saves, MIDI, every style and every line

ASan a stress test with AddressSanitizer and UBSan, random buttons and MIDI

Bugs the tests caught

- A buffer overflow in the stress test when copying tracks from the old format
- Percussion sounds 3.5 dB above the rest
- The summed groove going over full scale
- A FEEL value used as a hidden marker
- The editor sending the factory kit without the GROOVE bytes

ON THE DEVICE

The real FM-1, from Windows

Test	Result
Install over the factory firmware (V15)	✓ FM-1_923
Install over GR!MM	✓ FM-1_923
Install interrupted midway	✓ resumed by the loader
Sounds, GROOVE, lines, AMBIENTE	✓ by ear
Logged faults	✓ 0

CPU in the heaviest scene

47.3%

THE PROCESS

Seven phases in two days, paired with an AI agent

0	1	2	3	4	5	6
Setup and GR!MM build	Identity, saves, boot	PERC	GROOVE, hub, emulator	Lines and Latin sounds	Mood	Release and FM-1 test

The agent

Claude Code read the fork, wrote the C, the tests, the hub and the docs.

The human

Decided each phase's direction, did the pushes, listened and tested on the

The net

A plan written in phases and the tests as the definition of done.

LESSONS

What I’d take to the next project

1. Start from a living, well-tested fork. GR!MM's tests were the biggest inheritance.
2. Test away from the hardware. Host and WebAssembly caught almost everything.
3. Measure before you design. The full memory decided what AMBIENTE would be.
4. Data as text. Rhythms as strings: readable, reviewable, cheap.

THE TOOL

The method became a plugin that builds the next firmware

The way COQUÍ was built — the phases, the AC791N chip rules, the toolchain and the release — was packaged into a Claude Code plugin, **synth-firmware**, so the next FM-1 firmware reuses it all.

skill

The playbook: the phases, the chip rules, the toolchain and the release. It loads on its own.

agent

firmware-builder: takes a phase, builds it, respects the rules and runs the tests.

commands

`/firmware-new`
`/firmware-phase`
`/firmware-check`
`/firmware-release`

hook

A lint of the chip rules on every edit: float, 64-bit division, ROM-range constants.

Que siga la rumba

Hub and emulator • saulofilho.github.io/coqui-fm1

Code • github.com/saulofilho/coqui-fm1

Release • COQUÍ 1.0, GPL-3.0

Beginner's guide • PDF (EN/PT) on the hub

The tool • github.com/saulofilho/synth-firmware

Questions?

